AP PHYSICS 1

Unit 1 • Worksheet 1.5A

Worksheet 1.5A • Vectors and Motion in Two Dimensions

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A ball is launched at $25\,\text{m/s}$ at 37° above the horizontal ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$). What is the horizontal component of its velocity?

(A) $15\,\text{m/s}$

(B) $20\,\text{m/s}$

(C) $25\,\text{m/s}$

(D) $12.5\,\text{m/s}$

Question 2

A vector has components $A_x = 6$ and $A_y = -8$. What is its magnitude?

(A) 14

(B) 2

(C) 10

(D) 48

Question 3

A ball is thrown horizontally at $8\,\text{m/s}$ from the top of a cliff and lands $2\,\text{s}$ later. How high is the cliff?

(A) $20\,\text{m}$

(B) $16\,\text{m}$

(C) $10\,\text{m}$

(D) $40\,\text{m}$

Question 4

How far from the base of the cliff does the ball in the previous question land?

(A) $8\,\text{m}$

(B) $4\,\text{m}$

(C) $32\,\text{m}$

(D) $16\,\text{m}$

Question 5

At the same moment and from the same height, ball 1 is dropped and ball 2 is thrown horizontally. Which lands first?

(A) Ball 1
(B) Ball 2
(C) They land at the same time.
(D) It depends on their masses.

Question 6

What is the acceleration of a projectile at the highest point of its path?

(A) Zero
(B) g downward
(C) g upward
(D) Horizontal

Question 7

A ball is launched at $30\,\text{m/s}$ at 53° above level ground ($\cos 53^\circ = 0.6$, $\sin 53^\circ = 0.8$). How long is it in the air?

(A) $2.4\,\text{s}$
(B) $4.8\,\text{s}$
(C) $3\,\text{s}$
(D) $6\,\text{s}$

Question 8

Projectiles are launched with the same speed from level ground. Which launch angle gives the greatest range?

(A) 30°
(B) 60°
(C) 45°
(D) 90°

Question 9 (Free response)

A soccer ball is kicked from the ground at $20\,\text{m/s}$ at 37° above the horizontal ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$).

(a) Find the horizontal and vertical components of the initial velocity.
(b) Find the time to reach the highest point and the maximum height.
(c) Find the total time in the air and the horizontal range.
(d) What is the speed of the ball at its highest point?